CCNA 200-301 V2.0

# 02 The Physical Layer and Cable Faults

Part I — Foundations

EXAM TOPICS

1.1

LECTURER

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LEARNING OUTCOMES

# What you will be able to do

- **Select** the right medium — copper, multimode fibre, single-mode fibre — for a given distance and speed.
- **Read** the interface counters that expose a physical fault, and say which counter implicates which cause.
- **Diagnose** the classic faults: duplex mismatch, speed mismatch, wrong cable type, bad pin-out, dirty or over-long fibre, and a transceiver operating outside its optical budget.
- **Distinguish** a port that is down from a port that is up and damaging traffic — the second is far more dangerous.

ASSUMED KNOWLEDGE

# Before we start

Chapter 1 for the layered model and the idea of a collision domain.

KEY TERMS

# Vocabulary for this chapter

- twisted pair
- Cat5e/Cat6
- multimode
- single-mode
- SFP
- attenuation
- auto-negotiation
- duplex mismatch
- CRC error
- runt
- giant
- late collision
- input error
- pin-out
- straight-through
- crossover
- Auto-MDIX

Each is defined where it first matters — not here.

ON THE EXAM

# Read the topic wording

**This is what the blueprint actually asks for**

Topic **1.1** says *diagnose* interface and cable issues, and then names them: “collisions, errors, mismatched duplex, speed, distance, interface, signal levels, pin out, and cable types”. That list is a syllabus in itself. You will not be asked to describe a cable; you will be shown counters and asked what is wrong.

DEFINITION

# Bits per second, and why it is not bytes

Link speeds are always quoted in **bits** per second: 1 Gbps is one thousand million *bits* each second. File sizes are quoted in **bytes**, and a byte is eight bits.

So a 1 Gbps link moves at most about 125 megabytes per second, not 1000 — divide by eight. The lower-case `b` means bits and the upper-case `B` means bytes, which is the entire difference between `Mbps` and `MBps`.

Two further distinctions worth having straight from the start:

- **Bandwidth** is the rated capacity of the link — what it could carry.
- **Throughput** is what you actually got, after overhead, contention, errors and whatever the far end could manage.

Throughput is always lower, and a user reporting “slow” is reporting throughput. Nothing in this chapter changes bandwidth; several things in it quietly destroy throughput.

DEFINITION

# Duplex, and attenuation

**Duplex** is who may transmit, and when.

- **Half duplex** — one end at a time, like a walkie-talkie. If both transmit at once the signals overlap and the frames are destroyed; that overlap is a **collision**, and each end is expected to detect it, back off and retry.

DEFINITION

# Duplex, and attenuation (continued)

- **Full duplex** — both ends at once, on separate pairs of wires, like a telephone call. **Collisions are impossible**, so a full-duplex port that reports collisions is telling you something is badly wrong. That is the single most useful sentence in this chapter.

Modern switched links are full duplex, and both ends normally agree on it through **auto-negotiation** — a brief exchange when the link comes up in which each end states what it can do and both pick the best they share.

**Attenuation** is separate and simpler: a signal loses strength with distance and through material. Every distance limit in the table below is really an attenuation limit — the point at which the far end can no longer reliably tell a 1 from a 0.

CONCEPT

# Distance is a fault, not a preference

The exam names *distance* as a diagnosable issue for a reason. Exceed the reach of a medium and the link does not fail cleanly — it comes up and then corrupts frames, which shows as climbing CRC errors rather than a down port. A 95 m run that works in winter and fails in summer is a real phenomenon, and the answer is fibre, not another patch lead.

VERIFY

# show interfaces GigabitEthernet0/1

**SW1# show interfaces GigabitEthernet0/1**

```
GigabitEthernet0/1 is up, line protocol is up (connected)
  Hardware is Gigabit Ethernet, address is 00d0.58a1.0201
  MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec
  Full-duplex, 1000Mb/s, media type is 10/100/1000BaseTX
  5 minute input rate 41000 bits/sec, 12 packets/sec
     284517 packets input, 39281044 bytes, 0 no buffer
     Received 1204 broadcasts (0 multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     271033 packets output, 33771201 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 late collision, 0 deferred
```

CONCEPT

# The duplex mismatch, in one paragraph

Auto-negotiation lets two ports agree on speed and duplex. If one end is set to `duplex full` by hand and the other is left on auto, the auto end cannot detect duplex, falls back to half, and you have a mismatch. The half-duplex end listens before transmitting and treats simultaneous transmission as a collision; the full-duplex end transmits whenever it likes. The link stays *up*. It just loses a percentage of every conversation, and the symptom the user reports is “the network is slow”, not “the network is down”.

The fix is to set both ends the same way. The rule of thumb is: leave both on auto, or hard-code both. Never one of each.

CONFIGURE

# A hard-coded access port, both ends matching

Configure

```
interface GigabitEthernet0/1
 description Link to SW2 - speed and duplex fixed at both ends
 speed 1000
 duplex full
 no shutdown
```

VERIFY

# show interfaces GigabitEthernet0/25 transceiver

```
SW1#  show interfaces GigabitEthernet0/25 transceiver

                                   Optical     Optical
             Temperature  Voltage  Tx Power   Rx Power
Port         (Celsius)    (Volts)  (dBm)      (dBm)
---------    -----------  -------  --------   --------
Gi0/25         33.2         3.31     -2.4      -28.9
```

# Fibre and signal levels

- **Replacing the cable first.** It is the most expensive check in time and the least likely single cause on a link that has been working. Read the counters first: they tell you which of six causes it is.
- **Treating "up/up" as healthy.** A port can be up, passing traffic, and corrupting five per cent of it. The exam will show you exactly this and expect you not to declare it fine.
- **Hard-coding one end.** The commonest way to create a duplex mismatch is to fix one side "to be safe".
- **Reading raw totals instead of a rate.** Counters accumulate since the last clear. Ten thousand CRC errors over two years is noise; ten thousand since yesterday is a fault. Use `clear counters` and watch.

# “The file server is slow, but it is not down”.

A user copies a large file and gets a fraction of the expected throughput. Ping succeeds with no loss. You look at the access port:

# “The file server is slow, but it is not down”.

**What would you check first — and what do you expect to see?**

# show interfaces GigabitEthernet0/7 | include duplex|collision|CRC

**SW1# show interfaces GigabitEthernet0/7 | include duplex|collision|CRC**

```
  Half-duplex, 100Mb/s, media type is 10/100/1000BaseTX
     14432 input errors, 118 CRC, 0 frame, 0 overrun, 0 ignored
     0 output errors, 89211 collisions, 0 interface resets
     2244 late collision, 0 deferred
```

## What is the fault — and what does this output rule out?

# Why it is doing that

**Diagnosis.** The port is half duplex at 100 Mbps on gigabit-capable media, with a very large collision count and — decisively — *late* collisions. Late collisions on a modern switched link are the signature of a duplex mismatch: the far end is transmitting full duplex while this end still listens before sending. The small CRC count is a consequence of the collisions, not a separate cable fault.

# And the lesson

**Fix.** Find the server NIC setting. It has almost certainly been hard-coded to full duplex while the switch port was left on auto. Set both ends to auto, or both to `speed 1000` / `duplex full`, then `clear counters` and confirm the errors stop climbing. If they do not, *now* suspect the cable.

# Reading a broken link

**Topology.** Two 2960 switches connected by one link; a PC on each.

1. Cable `SW1 Gi0/1` to `SW2 Gi0/1`. Confirm both PCs can ping each other, then run `show interfaces Gi0/1` on both and record the speed, duplex and every error counter while the link is healthy.
2. On `SW1` only, set `speed 100` and `duplex half`. Leave `SW2` on auto.
3. Generate traffic between the PCs. Record what changes on each side. Which counter moved first?

# Reading a broken link (continued)

1. Repeat with `SW1` at `duplex full` and `SW2` on auto — the classic mismatch. Note that the link stays up.
2. **Extension.** Without looking at your own configuration, hand the topology to a partner, have them introduce one fault from this chapter, and diagnose it from `show` output alone in under three minutes.

**Verification.** You should be able to state, from counters only, whether a link is healthy, mismatched, or physically damaged.

A gigabit link is up. `collisions` is increasing and the interface reports full duplex. What single conclusion can you draw, and why is it certain?

ANSWER

# Checkpoint 1 of 3

**A gigabit link is up. `collisions` is increasing and the interface reports full duplex. What single conclusion can you draw, and why is it certain?**

> That the two ends disagree about duplex. It is certain because a full duplex link *cannot* have collisions — there is no shared medium to collide on. The counter can only be incrementing because the local end is actually running half duplex, or because the far end is and is causing them.

CHECKPOINT 2 of 3

## A fibre link shows Rx power of -31 dBm against a sensitivity of -23 dBm. Give three causes in the order you would check them.

Discuss · then advance

ANSWER

# Checkpoint 2 of 3

**A fibre link shows Rx power of -31 dBm against a sensitivity of -23 dBm. Give three causes in the order you would check them.**

> In order: a dirty or damaged connector; the wrong optic or wavelength for the distance; and excessive length or a bad splice. Check the cheapest first — clean the connector before ordering hardware.

CHECKPOINT 3 of 3

# Why is a port that is administratively down easier to deal with than one that is up with rising CRC errors?

Discuss · then advance

ANSWER

# Checkpoint 3 of 3

## Why is a port that is administratively down easier to deal with than one that is up with rising CRC errors?

> Because it is unambiguous. Administratively down is a decision somebody made, recorded in the configuration, and reversible with one command. A port that is up with rising CRC errors is intermittently working, so traffic is partly passing, users report “slow” rather than ”down”, and the cause could be cable, connector, optic, EMI or duplex.

# What to take away

- Copper is 100 m, full stop. Multimode reaches a campus; single-mode reaches a city. Exceeding a medium's reach corrupts frames rather than dropping the link.
- Diagnose from counters, not from intuition. CRC without collisions points at the cable; collisions on a full-duplex link point at duplex; late collisions are close to proof of a duplex mismatch.
- Set both ends of a link the same way — both auto, or both fixed.
- “Up/up” does not mean healthy. The dangerous fault is the one that passes most traffic and damages the rest.

# What to take away (continued)

- On fibre, read `show interfaces …{} transceiver`. Light levels are a diagnosable fault the blueprint names explicitly.

# Where we got to

- Copper is 100 m, full stop. Multimode reaches a campus; single-mode reaches a city. Exceeding a medium's reach corrupts frames rather than dropping the link.
- Diagnose from counters, not from intuition. CRC without collisions points at the cable; collisions on a full-duplex link point at duplex; late collisions are close to proof of a duplex mismatch.
- Set both ends of a link the same way — both auto, or both fixed.
- “Up/up” does not mean healthy. The dangerous fault is the one that passes most traffic and damages the rest.

NEXT

**Chapter 3 — Ethernet, Frames and Switching**